

Honghui Zhang

Intent World Models | Proactive LLM Agents | Multimodal Reasoning

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Education

Sichuan University (985) | Computer Science and Technology

2023.09 to Present

B.Eng., expected 2027 | Weighted Average Mark: 86.99/100

Research Interests: Intent modeling; proactive reasoning and decision making in LLM agents; multimodal LLMs; RAG and GraphRAG; preference alignment and reinforcement learning.

Manuscript Submitted

Honghui Zhang, Chenmeinian Guo, Yichen Yu, Guanyu Liu, Yujia Zhang, Yongming Qin, Chongguo Song, Mengyue Yang, Lei Yu, and Tianyu Shi. (2026). *See, Infer, Intervene: Proactive World Modeling for Goal-Oriented Social Intelligence*. **Submitted to NeurIPS**. arXiv:2606.03371.

Research Experience

Proactive Intent World Model (PIWM)

First Author / Lead Developer 2026.03 to Present

Contribution: Proposed PIWM and the See-Infer-Intervene framework to model customer states, salesperson actions, and subsequent responses. Designed training with asymmetric rewards and negative sampling for timely interventions, clarification questions, and product recommendations.

Outcome: Built GuidanceSalesBench with interaction outcomes judged by professional salespeople. Achieved a 40% improvement in sales decision optimization, with gains in customer state estimation, intent understanding, and outcome prediction.

DisCo: Joint Discrete and Continuous Molecular Diffusion

Lead Developer 2025.09 to Present

Contribution: Designed and implemented a joint diffusion model for discrete molecular features (atom types and bonds) and continuous 3D coordinates. Added direct preference optimization (DPO) to guide generation toward target molecular properties.

Outcome: Built a working prototype and stable training pipeline, increasing molecular validity from 30% to 50%. Gains from preference optimization remain limited and unstable; debugging is ongoing.

The Hong Kong University of Science and Technology

Research Assistant 2025.10 to 2026.02

Project: AI Tendering and Bidding System

Contribution: Led the initial system architecture and built a LangGraph pipeline with multiple agents to parse long tender documents, retrieve relevant content, and trace evidence to its source. Implemented document chunking and behavior tracking.

Outcome: Deployed for initial testing as the core tendering system for a major Hong Kong real estate enterprise, supporting multiple users, document management, and automated evaluation.

GraphRAG Regulatory Compliance Review System

Project Lead 2024.11 to 2025.11

Contribution: Led an Undergraduate Innovation Program project. Built an updatable knowledge graph from regulations, cases, and industry standards. Combined exact keyword search with semantic vector retrieval over the graph. Used OCR and a split and merge strategy to structure floor plans, tables, and text.

Outcome: Developed an automated occupational health assessment system that processes inspection reports of up to 200,000 characters and produces regulatory assessments. Filed one invention patent and one software copyright.

Teaching Experience

SCU-Pittsburgh Institute, Sichuan University

Data Analysis

Teaching Assistant

Computer Vision

Teaching Assistant

Awards & Competitions

China Undergraduate Mathematical Contest in Modeling (CUMCM), 2025: Provincial Second Prize. Implemented spatiotemporal optimization, 3D occlusion checks, and model pruning.

Tencent Kaiwu Honor of Kings Agent Competition, 2024: 4th place, Sichuan Province. Trained a 1v1 agent using KaiwuDRL and PPO, with reward shaping and evaluation across multiple heroes.

Skills

Programming & Frameworks: Python, PyTorch, FastAPI, LangGraph, Docker, Git